

FREIGHT RATE PREDICTION

Machine Learning Engineering Report

Spotter - Machine Learning Engineer Assessment

Hamzah Al-Basyouni | October 2026

48,000	12,000	\$98.49	4.14%
Labeled development loads	Future loads predicted	Average forward-validation MAE	Average forward-validation WAPE

Final approach A 75/25 CatBoost ensemble combining a log-transformed rate-per-mile model with a direct raw-rate model, selected using forward time-based validation.

1. Executive summary

The assessment is a forecasting-style tabular regression problem: 48,000 labeled loads span January through October 2025, while the 12,000 final prediction rows occur in November and December 2025. The modeling process therefore prioritized temporal generalization rather than random train/test accuracy.

The final solution uses two CatBoost models. The primary model predicts $\log_{1p}(\text{posted_rate} / \text{distance})$; the secondary model predicts `posted_rate` directly. A 75% / 25% blend produced the best average forward-validation MAE of \$98.49 and WAPE of 4.14%, compared with a \$233.25 MAE business baseline.

2. Data and validation design

2.1 Dataset structure

Dataset	Rows	Period	Target
Development	48,000	2025-01-01 to 2025-10-31	posted_rate
Final validation	12,000	2025-11-01 to 2025-12-31	Hidden

Because the final set is strictly later in time than the labeled data, random splitting would leak the deployment pattern into model selection. I used expanding-window forward validation:

Training window	Validation month
Jan-Jul 2025	August 2025
Jan-Aug 2025	September 2025
Jan-Sep 2025	October 2025

Why this matters Each validation fold simulates the real task: train on the past and predict a genuinely unseen future period.

2.2 Data-quality findings

- 300 missing training weights and 374 missing training market_index values.
- 292 negative training weights; these were converted to absolute magnitude and flagged with weight_was_negative.
- Eight pickup/delivery cities appear only in the final validation period.
- Approximately 1.4% of observations show extreme target residual behavior.
- quote_signal changes sign and strength materially across months, indicating temporal instability.

Month	Correlation	Month	Correlation
Jan	-0.600	Jun	-0.580
Feb	-0.639	Jul	+0.603
Mar	-0.595	Aug	+0.015
Apr	+0.613	Sep	-0.591
May	+0.624	Oct	+0.605

Decision: quote_signal was excluded from the selected production models because its relationship with the target was not stable over time.

3. Modeling approach

3.1 Feature engineering

- Route and categorical features: pickup, delivery, equipment, and pickup->delivery lane.
- Geographic features: coordinates, latitude/longitude deltas, and a geographic-distance proxy.
- Freight features: corrected weight, log-weight, distance, log-distance, and weight-per-mile.
- Calendar features: month, week, day-of-week, day-of-year, plus cyclical sine/cosine encodings.
- Missing-value indicators for weight and market_index.

CatBoost was chosen because it handles mixed numeric/categorical tabular data well, supports missing numeric values, and can use geographic features as a fallback for unseen city categories.

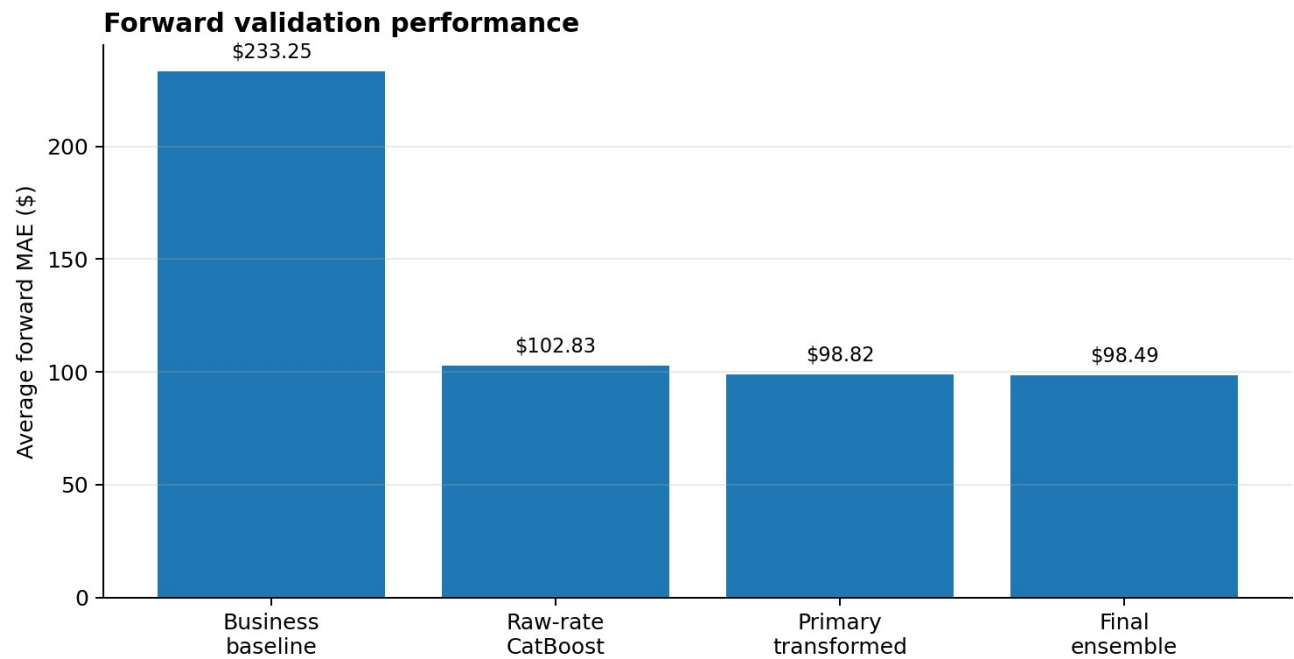
3.2 Target and model selection

Target representation	Avg MAE	Avg WAPE	Median AE	P95 AE
raw_rate	\$102.82	4.33%	\$33.98	\$167.47
log_rate	\$108.57	4.56%	\$34.05	\$197.63
rate_per_mile	\$99.56	4.19%	\$30.30	\$161.68
log_rate_per_mile	\$99.22	4.17%	\$30.39	\$161.10

The transformed target was tuned with a deliberately small search due to the assessment time limit. The best stable configuration used depth 7, learning rate 0.05, L2 regularization 10, and random strength 0.5.

Outlier decision Cross-fitted screening consistently identified about 1.38% extreme labels. Removing them improved average MAE only from \$102.83 to \$101.88 and worsened September slightly, so all labeled rows were retained and MAE loss was used for robustness.

4. Validation results



Model	Avg MAE	Avg RMSE	WAPE	Median AE	P95 AE
Business baseline	\$233.25	\$669.52	9.83%	\$114.14	-
Raw-rate CatBoost	\$102.83	\$626.66	4.33%	\$34.07	-
Primary transformed	\$98.82	\$626.12	4.16%	\$30.23	\$160.07
Final 75/25 ensemble	\$98.49	\$626.05	4.14%	\$29.85	\$158.55

The final ensemble reduced average MAE by approximately 57.8% relative to the equipment-specific median rate-per-mile baseline. The remaining RMSE is dominated by a small set of extreme target observations: in out-of-time residual analysis, non-extreme rows had an MAE of \$48.87 while the extreme subset averaged \$3,679.39 absolute error.

4.1 Feature importance

Transformed model	Importance	Raw-rate model	Importance
weight_per_mile	27.724	distance_log	38.264
equipment	27.017	distance	27.880
geo_distance_proxy	9.027	geo_distance_proxy	14.627
distance	7.180	lon_delta	3.570
distance_log	4.420	equipment	3.480

5. Final predictions and sanity checks

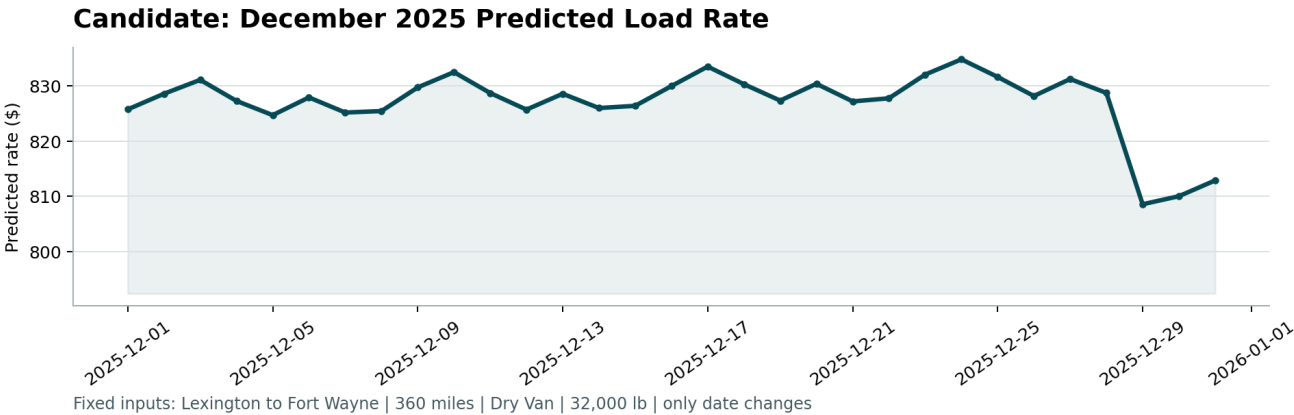
5.1 Final 12,000-load output

Statistic	Training actual rate	Final predicted rate
Minimum	\$57.22	\$192.69
P01	\$327.17	\$342.74
Median	\$2,030.76	\$2,028.00
Mean	\$2,373.98	\$2,345.77
P95	\$4,953.77	\$4,897.07
P99	\$5,972.83	\$5,723.06
Maximum	\$25,533.00	\$6,707.43

The central distribution of final predictions closely matches the labeled data, while the very large anomalous training tail is not reproduced. Predicted rate-per-mile also remains close to training: median 2.135 versus 2.145 historically.

5.2 Fixed December scenario

The provided chart scenario fixes Lexington -> Fort Wayne, 360 miles, Dry Van, and 32,000 lb; only the date changes. Fields not supplied by the December input file (for example market_index) are left missing rather than inferred from the final validation set.



6. December sanity check

Reference	Rows	Rate / summary
Exact Lexington -> Fort Wayne Dry Van history	21	Median \$807.89; median RPM \$2.233
Comparable 300-420 mile, 28k-36k lb Dry Van history	755	Median \$844.33; median RPM \$2.35
December model scenario	31	Mean \$827.00; range \$808.56-\$834.76

The December series sits between the exact-lane historical median and the broader comparable short-haul Dry Van median, which is a reasonable sanity check for the fixed scenario.

7. Final training and reproducibility

- Random seed: 42.
- Final iteration counts calibrated on October 2025, the latest labeled month.
- Transformed model: 270 iterations after recent-period calibration.
- Raw-rate model: 867 iterations after recent-period calibration.
- Both final models retrained on all 48,000 labeled January-October rows.
- Final predictions saved as `validation_predictions.csv` using the official `load_id` template.
- The provided `score.py` successfully validated all 12,000 predictions and all 31 December rows and generated `scorer_results/candidate_december.png`.

8. Conclusion

The final solution emphasizes temporal validity, robust handling of noisy labels, and reproducibility. The selected ensemble materially outperforms a freight-domain baseline while avoiding features that showed unstable behavior across time. The final output passed the provided structural scorer; hidden final validation metrics are calculated by Spotter after submission.